Gain-weighted well-counting in a congestion mean-field game

Certified equilibria whose density carries more local maxima than the potential has wells — two theorems, a bracket table of certified instances, and an exact, gamma-independent crossover constant.

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Abstract

We prove, by validated numerics, that a stationary discounted congestion mean-field game on the torus admits equilibria whose population density has strictly more local maxima than the cost potential has wells. With $V(x) = A1 \cos 2\pi x + A2 \cos 4\pi x$ and $A1 > 4A2$ — exactly ONE well — the certified equilibrium density at our main instance has EXACTLY 2 strict local maxima (Theorem 1); a third-harmonic instance yields EXACTLY 3 (Theorem 2). The honest mechanism is elementary and we state it as such: the equilibrium re-weights harmonics **the potential already contains** through the band-pass linear-response gain $c(\kappa) = \kappa/[(1+\sigma\kappa)^2 + \gamma\kappa]$, so the density counts wells with gain-weighted amplitudes while the potential counts them flat. The second-vs-first harmonic crossover σ *= 1/(8π²) is INDEPENDENT of the coupling γ — decided here as an exact-rational polynomial identity (the γ-coefficient of the crossover polynomial is identically zero), not by float agreement — and the limiting splitting window is the exact rational interval (1/16, 1/4) (third harmonic: (1/27, 1/3), the Chebyshev U_{k−1} law). Around each theorem sit five further certified instances forming a bracket table: negatives below the amplitude threshold and above σ*, replications, and a threshold pin — the splitting threshold at $\sigma = 0.002$, $\gamma = 0.01$ lies in [0.13, 0.14] by certified counts, with the exact-rational linear-response prediction $r_c = 0.132725$ inside. All enclosures are radii-polynomial certificates at radius ~2.52e-13 with local uniqueness in the full sequence-space ball (even AND odd blocks) and both positivity walls certified. This document is generated from those certificates.

1. Introduction

Under Lasry–Lions monotonicity, mean-field-game theory guarantees a unique equilibrium, and the intuition that travels with the theory is stronger than the theorem: the population's density is expected to mirror the cost — wells of the potential become peaks of the crowd, one for one. The known rigorous results reinforce that reading (in the Cesaroni–Cirant regime, density peaks sit where the potential's wells sit), and the natural working assumption is a CEILING: the equilibrium cannot carry more local maxima than the cost has wells.

This paper proves the ceiling false, inside the uniqueness regime, by validated numerics. At explicit parameters of a stationary discounted congestion mean-field game on the torus we certify exact equilibria whose density carries EXACTLY two — and, at a third-harmonic instance, EXACTLY three — strict local maxima over a potential with EXACTLY one well. The mechanism is not exotic and we state it at its true altitude: the linearized response to the cost is a band-pass filter in harmonic frequency, so the equilibrium counts wells with GAIN-WEIGHTED amplitudes while the potential counts them flat, and between those two counts a single well splits the crowd. What the linear analysis cannot give —

and the certificates do — is the nonlinear statement: these are exact equilibria of the full congestion system, with local uniqueness in the full sequence-space ball and both positivity walls certified.

Contributions. (i) The phenomenon theorem and its three-peak companion (Sections 4–5), by radii-polynomial enclosures whose certified balls fix the exact critical-point count of the exact solution. (ii) The crossover constant sigma* = 1/(8 pi^2), INDEPENDENT of the discount, decided as an exact-rational polynomial identity rather than observed numerically (Section 3). (iii) A bracket table of certified instances pinning the splitting threshold inside [0.13, 0.14], with the exact-rational linear-response prediction landing inside the pin (Section 6). (iv) An exact solution count for the companion ergodic system — EXACTLY three Galerkin solutions, box-bounded — together with function-space multiplicity: at least three distinct exact solutions per coupling in pairwise disjoint uniqueness balls (Section 7). Every number in this document is interpolated at build time from machine-written certificates (named where used); the build refuses to render if any certificate is missing, refused, or structurally moved.

2. The model and the exact instances

Stationary discounted congestion MFG on the unit torus (congestion exponent a = 1/2):

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u - σu″ + ½ (u′)²/√m = V(x) + γ m
m - σm″ - (√m · u′)′ = 1,   m > 0,
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$V(x) = A1 \cos 2\pi x + A2 \cos 4\pi x + A3 \cos 6\pi x$, augmented with the branch variable $w = m^{-1/2}$ and the reciprocal constraint $m \cdot w^2 = 1$, solved in even cosine Fourier space with the derivative field $p = u'$ carried odd.

Coefficient honesty. Each theorem is a statement about specific binary64 doubles, not about the decimal strings that name them. Main instance (T1): σ = 0.002 (0x1.0624dd2f1a9fcp-9), γ = 0.01 (0x1.47ae147ae147bp-7), A1 = 0.003 (0x1.89374bc6a7efap-9), A2 = 0.0006000000000000001 (0x1.3a92a30553262p-11), N = 96, ν = 1.02. Three-peak instance (T6): A3 = 0.00075 (0x1.89374bc6a7efap-11), A2 = 0, same σ, γ, N, ν.

3. Linear response: the band-pass gain and the exact crossover

At first order in the potential amplitudes the congestion term drops and each harmonic responds independently: $\hat{m}_k = -c(\kappa_k) A_k$ with $\kappa_k = (2\pi k)^2$ and

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c(κ) = κ / [ (1 + σκ)² + γκ ].                              (†)
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Three facts about (†) are decided in EXACT RATIONALS (`certs/terra-sigmastar.json`; stdlib fractions, no floats in any verdict):

Proposition 1 (band-pass). As polynomials in Q[σ, κ, γ]: $D - \kappa\, \partial D/\partial\kappa = (1 + \sigma\kappa)(1 - \sigma\kappa)$, where D is c's denominator. The γ-term cancels identically, so $\operatorname{sign}(dc/d\kappa) = \operatorname{sign}(1 - \sigma\kappa)$ for every γ: the gain peaks at $\kappa = 1/\sigma$.

Proposition 2 (γ-free crossover). With $s = \sigma\kappa_1$, $g = \gamma\kappa_1$, the k-th-vs-first crossover polynomial $N_k(s, g) = k^2[(1+s)^2 + g] - [(1+k^2 s)^2 + k^2 g]$ factors as $(k^2 - 1)(1 - k^2 s^2)$: its g-coefficient is the ZERO polynomial and its positive root is $s = 1/k$ exactly (decided for $k = 2..12$). Hence $c_k = c_1$ exactly at $\sigma = 1/(4\pi^2 k)$; at $k = 2$,

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σ* = 1/(8π²) = 0.012665147955292222…
```

(rational bracket of width 1.26e-44 via Machin's formula — the exact rational endpoints are in the certificate). The γ-independence is an exact cancellation, not numerical smallness.

Proposition 3 (the splitting windows). $\sin 2\pi kx = \sin 2\pi x \cdot U_{k-1}(\cos 2\pi x)$, so V acquires interior critical points beyond $x = 0, ½$ iff $k\cdot r\cdot|\min U_{k-1}| > 1$ ($r = A_k/A1$). The flat thresholds are exact rationals from the ranges of $U_1 = 2c$ and $U_2 = 4c^2 - 1$: threshold 1/4 at $k = 2$ and 1/3 at $k = 3$. The equilibrium obeys the SAME criterion with gain-weighted amplitudes, and $c_k/c_1 \to k^2$ as $(\sigma, \gamma) \to 0$, so the limiting splitting windows are exactly (1/16, 1/4) and (1/27, 1/3).

One sentence, stated honestly: between the gain-weighted count and the flat count, the crowd splits at the bottom of a single well. The mechanism — non-flat linear gain changing the critical-point count of a two-harmonic signal — is elementary [CITE NEEDED: a textbook-level reference; stating this as elementary isolates the actual contribution, which is the MFG setting, the exact γ-free constant, and the nonlinear certificates below]. Linear response PREDICTS; only the certificates PROVE.

4. Theorem 1 — two peaks over one well

Theorem 1. At the T1 instance of §2 (A1 > 4A2: V has exactly one well), there is an exact equilibrium (u, m, w) of the full congestion system within ℓ^1_v distance 2.520e-13 of the stored candidate, locally unique in the full sequence-space ball — even (cosine) AND odd (sine) blocks — with $m \geq 0.8944$ and $w \geq 0.9743$ over the WHOLE ball, and EVERY density in that ball has EXACTLY 2 strict local maxima and 2 strict local minima on the torus, while V has exactly 1 well. (`certs/terra-recert-t1.json`, `certs/terra-peakcount-t1.json`.)

Proof numbers (radii-polynomial / Newton–Kantorovich, all bounds in outward-rounded interval arithmetic): Y0 = 2.472e-14, Z1 = 0.8970 (even 0.8970 / odd 0.8970), Z2 = 1131.8243, closure margin 4.690e-6. The peak count derives ONLY from certified region signs: an alternating chain of curvature and slope regions covering [0, ½], each sign certified over the whole ball with the ball's derivative pads folded into every cell bound ($\sup|\delta m'| \leq$ 2.941e-11, $\sup|\delta m''| \leq$ 1.373e-8), with the smallest certified margin 6.72e-2 — more than six orders above the pads.

5. Theorem 2 — three peaks over one well

Theorem 2. At the T6 instance (third harmonic, A3/A1 = 0.25, one well), the analogous statement holds with radius 2.731e-13, $m \geq 0.8823$, and EXACTLY 3 strict local maxima (3 minima); V has exactly 1 well. (`certs/terra-recert-t6.json`, `certs/terra-peakcount-t6.json`.)

6. The bracket table — the certified instances under one theorem

Honest counting: the finding is the two theorems above plus THIS TABLE — rows of a table are rows of a table, not further theorems.

instance	σ	ratio	N	radius r	Z1	min m	peaks / wells
T1	0.002	A2/A1 = 0.2	96	2.52e-13	0.8970	0.8944	2 / 1
T2	0.002	A2/A1 = 0.12	96	2.25e-13	0.8889	0.9059	1 / 1
T3	0.02	A2/A1 = 0.2	64	2.97e-14	0.5235	0.9621	1 / 1
T4	0.002	A2/A1 = 0.15	96	2.37e-13	0.8919	0.9016	2 / 1
T6	0.002	A3/A1 = 0.25	96	2.73e-13	0.9046	0.8823	3 / 1
T7	0.002	A2/A1 = 0.13	96	2.30e-13	0.8899	0.9044	1 / 1
T8	0.002	A2/A1 = 0.14	96	2.32e-13	0.8909	0.9030	2 / 1

T2 ($r = 0.12$) and T3 ($\sigma = 0.02 > \sigma^*$) are one-peak negatives exactly where the linear response predicts none; T4 replicates the split at $r = 0.15$; T7/T8 pin the splitting threshold at $\sigma = 0.002$, $\gamma = 0.01$ inside **[0.13, 0.14]** by certified counts, and the exact-rational prediction $r_c = [(1+4s)^2 + 4g]/(16[(1+s)^2 + g]) = 0.132725\ldots$ (rational bracket in `certs/terra-bracket-table.json`) lands inside the pin.

7. Companion results — an exact count, and certified multiplicity

For the ergodic mfg-cap system ($V \equiv 0$, monotone coupling $c = -12$, $\sigma = 1/2$), a Krawczyk exhaustion census proves the N-mode even Galerkin truncation has EXACTLY 3 solutions in an explicit printed box for N = 2, 3, 4, 5 (N = 5: 6,954,073 boxes; every subbox eliminated by interval-residual or Krawczyk exclusion, each solution isolated by Moore–Krawczyk $K(X) \subset \mathrm{int}(X)$). The claim is stated box-bounded and truncation-level; the PDE-level count is an open problem. (`certs/mfg-cap-census-N{2..5}-c-12.json`.)

Its FUNCTION-SPACE companion: at each of six couplings c = -11, -12, -14, -16, -20, -24 — past the Lasry–Lions monotonicity wall $c^* = -\sigma^2(2\pi)^2$ — the constant solution, the symmetry-broken branch and its half-shift mirror are enclosed in PAIRWISE DISJOINT ℓ^1_v uniqueness balls with certified positive density (deepest floor min $m \geq 4.45\text{e-}4$ at $c = -24$): AT LEAST THREE distinct exact solutions of the SYSTEM at every listed coupling, and the half-shift symmetry provably produces a different solution, not a relabeling. At $c = -9.5$, inside the monotone regime, the branch collapses onto the constant and no claim is made. (`certs/mfg-cap-multiplicity.json`.)

8. Method and verification posture

The enclosures are radii-polynomial certificates on the augmented (a_0, p, m, w) system with the reciprocal constraint, ℓ^1_v Banach-algebra tail bounds (the enclosed object solves the SYSTEM, not an N-mode approximation), and both blocks of the linearization certified — the odd block is where a symmetry-breaking solution would live. Two independent implementations of the argument — a JavaScript kernel and the stdlib-Python verifier lineage that ships inside the companion page — agree on the certified T1 radius to the last digit, with independently computed approximate inverses on each side. The verifier's falsifier battery (nine controls per instance) includes two that attack the extension's own new lines: zeroing the instance's harmonic data term and moving it to the wrong mode must each explode the residual — and do. Positivity of m and of the branch selector w is certified over the whole ball, never observed on a plot. Peak counts never trust a float sign: floats propose, certified region chains decide, and anything uncertifiable REFUSES.

The falsified prior worth naming: the belief that equilibrium peaks cannot exceed potential wells was THIS LAB'S OWN working assumption, formed on an unreproduced ~508-sample float campaign — not folk belief, and not literature. The certificates above refute it.

9. Related work and fences

- Karuturi, arXiv:2605.20213 — THE MANDATORY FENCE: spontaneous instability of the uniform state under an interaction kernel with NO potential, with a cognate $8\pi^2\nu^2$ constant for a different quantity. Ours is the driven response to a structured cost under a fixed potential, inside the uniqueness regime.
- Cesaroni–Cirant, arXiv:1705.10741 — the foil: there, density peaks mirror the potential's wells; here they certifiably exceed them.
- Osborne–Smears, arXiv:2502.14687 — different in kind (FEM a-posteriori analysis, not equilibrium enclosures).
- Cecchin–Dai Pra–Fischer–Pelino, arXiv:1810.05492 — non-uniqueness via a different mechanism (finite state space).
- van den Berg–Lessard (radii polynomials) and Rump (Krawczyk / interval verification) — the CAP lineage this work instantiates. [OPERATOR/CITE: exact reference strings before submission.]

10. Open problems

PDE-level census (rung b); k-peak uniformity in k; sharper r_c digits; non-even branches as $\gamma \to 0$; the time-dependent forward–backward CAP.

Status and reproducibility

Nothing in this program has been published or sent anywhere; every send is released only by the author. Every certificate re-runs from the repository (`python3 instruments/mfgcap/run_recert.py t1`, `node instruments/critcount/run.js t1`, `python3 instruments/mfgcap/sigmastar.py`, `node labs/mfg/census.js`, `node labs/mfg/multiplicity.js`, `make test`). The base instance of this program — the $\sigma = 1/2$ single-harmonic enclosure of `reports/mfg-congest.html` — is the companion page whose embedded stdlib verifier serves as the calibration gate of every certificate here; the two are one program and any priority claim is made by their first release together.

Generated by tools/build-terra-writeup.js from the named certificates; the build refuses if any input is missing, refused, or moved. Draft v0.2 wants a human read [OPERATOR] before anything further; nothing is sent anywhere without explicit operator release.